Logics & Symbolic AI – cheat sheet

notation = course slides (Holzenberger, Simidjievski, Suchanek, Bonald, Bouzeghoub)

1 Propositional logic – syntax & semantics

Syntax (how to manipulate symbols) vs **semantics** (meaning). Alphabet: propositional symbols p; constants $T, \bot$; connectors $\neg$ (1-place), $\wedge, \vee, \ldots$ (2-place). Formula: atomic; if F formula then $(\neg F)$; if $\bullet$ connector, $(A \bullet B)$. Smallest such sets.

Syntactic $\bullet$ vs semantic $\blacksquare$: $\neg$/Not, $\wedge$/And, $\vee$/Or, $\supset$/$\Rightarrow$, $\subset$/$\Leftarrow$, $\equiv$/$\Leftrightarrow$, $\uparrow$/Nand, $\downarrow$/Nor, $\not\supset$/$\not\Rightarrow$, $\not\subset$/$\not\Leftarrow$.

Valuation $v : F \to \{True, False\}$, consistent: $v(T) = True$, $v(\bot) = False$, $v(\neg F) =$ Not $v(F)$, $v((A \bullet B)) = v(A) \blacksquare v(B)$.

$2^{2^2} = 16$ two-place connectors, 6 trivial (T, $\bot$, A, B, $\neg A$, $\neg B$); 3-place: $2^{2^3} = 256$.

Tautology: $v(X) = True$ for all v. **Satisfiable** set S: some v_0 makes every $X \in S$ True (SAT: find v_0). X tautology iff $\neg X$ not satisfiable.

Consequence $S \models X$: every v making all of S True makes X True; $\models X$ = tautology.

$S \models X$ **iff** $S \cup \{\neg X\}$ not satisfiable. ($\Rightarrow$: if v falsifies some $A \in S$ done, else $v(X) = True$ so $v(\neg X) = False$. $\Leftarrow$: v satisfies $S \Rightarrow v(\neg X) = False \Rightarrow v(X) = True$.)

Ex falso: $A, \neg A \in S \Rightarrow \forall X, S \models X$. Monotony: $S \models X \Rightarrow S \cup \{A\} \models X$. Deduction: $S \cup \{X\} \models Y$ iff $S \models (X \supset Y)$. X taut. iff $T \supset X$ taut. ($v(T \supset X) = v(X)$).

Laws: $\neg(X \wedge Y) \equiv \neg X \vee \neg Y$; $\neg(X \vee Y) \equiv \neg X \wedge \neg Y$; $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$; $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$.

2 Rewriting to CNF & resolution

Normal form for negation: $\neg$ only in front of symbols; use $\neg\neg X \equiv X$, De Morgan.

Generalized disjunction $[X_1, \ldots, X_n]$: False iff all X_i False; $v([]) = False$. **Generalized conjunction** $\langle X_1, \ldots, X_n \rangle$: True iff all True; $v(\langle\rangle) = True$.

CNF $\langle C_1, \ldots, C_n \rangle$, each **clause** $C_i = [X_1, \ldots, X_{n_i}]$ (DNF: mutatis mutandis).

α/β rewriting	negated
$\wedge$ $X \wedge Y$ (α)	$\neg(X \wedge Y) \equiv \neg X \vee \neg Y$
$\vee$ $X \vee Y$ (β)	$\neg(X \vee Y) \equiv \neg X \wedge \neg Y$
$\supset$ $X \supset Y \equiv \neg X \vee Y$	$\neg(X \supset Y) \equiv X \wedge \neg Y$
$\subset$ $X \subset Y \equiv X \vee \neg Y$	$\neg(X \subset Y) \equiv \neg X \wedge Y$
$\uparrow$ $X \uparrow Y \equiv \neg X \vee \neg Y$	$\neg(X \uparrow Y) \equiv X \wedge Y$
$\downarrow$ $X \downarrow Y \equiv \neg X \wedge \neg Y$	$\neg(X \downarrow Y) \equiv X \vee Y$
$\not\supset$ $X \not\supset Y \equiv X \wedge \neg Y$	$\neg(X \not\supset Y) \equiv \neg X \vee Y$
$\not\subset$ $X \not\subset Y \equiv \neg X \wedge Y$	$\neg(X \not\subset Y) \equiv X \vee \neg Y$

($\equiv$: $X \equiv Y$ as $(X \supset Y) \wedge (Y \supset X)$.)

Rewriting algorithm: disjunction $\langle \ldots [\ldots P \ldots] \ldots \rangle \to \langle \ldots [\ldots A, B \ldots] \ldots \rangle$; conjunction $\to \langle \ldots [\ldots A \ldots], [\ldots B \ldots] \ldots \rangle$; $\neg\neg P \to P$.

Ex. $((A \supset B) \supset (A \supset C))$: $\langle [\neg(A \supset B), (A \supset C)] \rangle \to \langle [A \wedge \neg B, A \supset C] \rangle \to \langle [A, \neg A, C], [\neg B, \neg A, C] \rangle$; 1st clause always True $\Rightarrow \langle [\neg B, \neg A, C] \rangle$.

Proof by resolution: sequence = conjunction of lines, each line a clause. $[\ldots(\beta_1 \vee \beta_2)\ldots] \Rightarrow$ add $[\ldots\beta_1, \beta_2\ldots]$; $[\ldots(\alpha_1 \wedge \alpha_2)\ldots] \Rightarrow$ add $[\ldots\alpha_1\ldots]$ and $[\ldots\alpha_2\ldots]$; replace $\neg\neg X$ by X, $\neg T$ by $\bot$, $\neg\bot$ by T. **Resolution**: from $[A, X, B]$, $[C, \neg X, D]$ create $[A, B, C, D]$. Proof of X: starts with $[\neg X]$ (goal), ends with []. To prove $S \vdash X$ start with $[S_1], \ldots, [S_n], [\neg X]$. X tautology iff it has a resolution proof.

Ex. $((A \supset B) \wedge (B \supset C)) \supset \neg(\neg C \wedge A)$: $[\neg(\ldots)] \to [(A \supset B) \wedge (B \supset C)], [\neg C \wedge A] \to [\neg A, B], [\neg B, C], [\neg C], [A] \to [B] \to [C] \to []$.

Prolog: `A :- B` becomes $[A, \neg B]$; query `?- g` adds $[\neg g]$.

Backward chaining (from goal; resolution, Robinson 1965) vs **forward chaining** (from assumptions; Logic Theorist, Newell 1956). Resolution can explode combinatorially.

Axiomatics. $\models$ semantic (truth tables), $\vdash$ syntactic (resolution). Hilbert system: $X \supset (Y \supset X)$; $(X \supset (Y \supset Z)) \supset ((X \supset Y) \supset (X \supset Z))$; $\bot \supset X$; $X \supset T$; $\neg\neg X \supset X$; $X \supset (\neg X \supset Y)$; $(A \wedge B) \supset A$; $(A \wedge B) \supset B$; $(A \supset X) \supset ((B \supset X) \supset ((A \vee B) \supset X))$; rule **modus ponens** $\frac{X \quad X \supset Y}{Y}$ (forward chaining).

Sound: $S \vdash F \Rightarrow S \models F$. **Complete**: $S \models F \Rightarrow S \vdash F$. Gödel (PhD 1929): resolution sound & complete for FOL.

Datalog (lab): rules $\forall\vec{x}\,(P_1(\vec{w}_1) \wedge \cdots \wedge P_k(\vec{w}_k)) \Rightarrow P(\vec{y})$, closed; fact $= T \Rightarrow P(\vec{c})$. **Minimal model property**; no negation (would lose it). Syntax: Upper = variable, lower = constant; `P(X,Z,alice) :- S(X,Y), T(Y,Z).`; query `R(X,charlie)?`. e.g. `grandparent(X,Z) :- parent(X,Y), parent(Y,Z).`

3 First-order (predicate) logic

Terms: variables x_i, constants c_i, functors $f(t_1, .., t_n)$ (f of arity n). Signature $= \{x_i\}, \{c_i\}, \{p_i\}, \{f_i\}$. **Atomic formula** $p(t_1, \ldots, t_n)$, p predicate of arity n. Formulas: atomic, $\neg F$, $(F \circ G)$, $(\forall x)F$, $(\exists x)F$. **Function vs predicate?** A symbol *inside* an argument position (a term) is a functor; a symbol applied at formula level (operand of $\neg, \wedge, \supset$, under a quantifier) is a predicate. Arity = #arguments. Constant = 0-ary functor.

Free/bound: term closed iff no variable. v free in atomic F iff occurs in F; free in $(\diamond v')F$ iff free in F and $v \neq v'$; free in $\neg F$ iff free in F; free in $F_1 \circ F_2$ iff free in F_1 or F_2. Bound = no free occurrence. Closed formula = no free variable. (Same name can be free in one place and bound in another!)

Semantics: non-empty domain D; interpretation I: constant $c \mapsto c^I \in D$, functor $f \mapsto f^I : D^n \to D$, predicate $p \mapsto p^I$ n-ary relation on D. Assignment A: variable $v \mapsto v^A \in D$. $t^{I,A}$: c^I, v^A, $f^I(t_1^{I,A}, \ldots)$. Model $M(D, I)$.

Attitude (truth value): $p(t_1..t_n)^{I,A} = True$ iff $(t_1^{I,A}, ..) \in p^I$; $((\forall x)F)^{I,A} = True$ iff $F^{I,B} = True$ for all B equal to A save for x; $\exists$: for at least one such B.

F **true in** M if True for all A; **valid** if true in every model. S satisfiable in M if some A makes all $F \in S$ True. F valid iff $\{\neg F\}$ not satisfiable.

Valid: $((\forall x)p(a, x)) \supset p(a, a)$; $(\forall x(p \wedge q)) \equiv ((\forall x)p \wedge (\forall x)q)$. Not valid, sat.: $(\forall x)(p(a, x) \supset p(a, a))$, $(\forall x)(\forall y)(p(x, y) \supset p(y, x))$, $(\forall x)(\exists y)p(x, y)$. Unsat: $\exists x(\neg p(x) \wedge p(x))$.

Prenex (rename variables first!): $\neg(\exists v)A \equiv (\forall v)\neg A$; $\neg(\forall v)A \equiv (\exists v)\neg A$; $((\forall v)A \wedge B) \equiv (\forall v)(A \wedge B)$ (same for $\exists$, either side); $((\forall v)A \supset B) \equiv (\exists v)(A \supset B)$; $(A \supset (\forall v)B) \equiv (\forall v)(A \supset B)$; $((\exists v)A \supset B) \equiv (\forall v)(A \supset B)$; $(A \supset (\exists v)B) \equiv (\exists v)(A \supset B)$. ($\forall$ in antecedent flips: $(\forall x)A \supset B \equiv \neg(\forall x)A \vee B \equiv (\exists x)(\neg A \vee B)$.)

Skolemization: $(Q_1 x_1)..(Q_{k-1}x_{k-1})(\exists x_k)(Q_{k+1}x_{k+1})..F \to (Q_1 x_1)..F(x_k/f(x_1, .., x_{k-1}))$; f new Skolem functor, arity $k-1$ (= #$\forall$ *before* it; arity 0 = constant). Preserves satisfiability. Put $\exists$ first if possible. E.g. $(\forall x)(married(x) \supset (\exists y)spouse(x, y)) \to (\forall x)(married(x) \supset spouse(x, f(x)))$.

Resolution procedure to prove F valid: 1 $\neg F$ to prenex (rename); 2 skolemize; 3 drop $\forall$; 4 CNF (opt.); 5 resolution + **unification** (e.g. $\neg P(c)$, $P(x)$: $x = c \Rightarrow$ []); 6 [] reached $\Rightarrow \{\neg F\}$ unsat $\Rightarrow F$ valid.

	Propositional	First-order
semantics	valuation, tautology	model, attitude, valid
proof	resolution	+ skolem, unification
	sound, complete,	sound, complete,
	decidable (SAT)	**semi-decidable**

Semi-decidable: if valid, a proof is found in finite time; if not valid, no guarantee to terminate. Gödel 1931: no complete method for arithmetic (recurrence).

4 Natural deduction (handout)

$$\text{Ax}\ \frac{}{\Gamma, p \vdash p} \qquad \neg\text{I}\ \frac{\Gamma, p \vdash \bot}{\Gamma \vdash \neg p}$$

$$\neg\text{E}\ \frac{\Gamma \vdash p \quad \Gamma \vdash \neg p}{\Gamma \vdash \bot} \qquad \text{Abs}\ \frac{\Gamma, \neg p \vdash \bot}{\Gamma \vdash p}$$

$$\bot\text{E}\ \frac{\Gamma \vdash \bot}{\Gamma \vdash p} \qquad \to\text{I}\ \frac{\Gamma, p \vdash q}{\Gamma \vdash p \to q}$$

$$\to\text{E}\ \frac{\Gamma \vdash p \to q \quad \Gamma \vdash p}{\Gamma \vdash q} \qquad \wedge\text{I}\ \frac{\Gamma \vdash p \quad \Gamma \vdash q}{\Gamma \vdash p \wedge q}$$

$$\wedge\text{Eg}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash p} \qquad \wedge\text{Ed}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash q}$$

$$\vee\text{Ig}\ \frac{\Gamma \vdash p}{\Gamma \vdash p \vee q} \qquad \vee\text{Id}\ \frac{\Gamma \vdash q}{\Gamma \vdash p \vee q}$$

$$\vee\text{E}\ \frac{\Gamma \vdash p \vee q \quad \Gamma, p \vdash r \quad \Gamma, q \vdash r}{\Gamma \vdash r} \qquad \top\text{:}\ \frac{}{\Gamma \vdash \top}$$

$$\forall\text{I}\ \frac{\Gamma \vdash F}{\Gamma \vdash \forall x F}\ x \text{ not free in } \Gamma \qquad \forall\text{E}\ \frac{\Gamma \vdash \forall x F(x)}{\Gamma \vdash F[t/x]}$$

$$\exists\text{I}\ \frac{\Gamma \vdash F(t)}{\Gamma \vdash \exists x F(x)} \qquad \exists\text{E}\ \frac{\Gamma \vdash \exists x F \quad \Gamma \vdash F \to G}{\Gamma \vdash G}$$

$\exists$E: x not free in Γ and G. Abs and $\bot$E: proof by contradiction / ex falso.

Ex. $\vdash (A \to B) \to ((B \to C) \to (A \to C))$. Let $\Gamma = \{A \to B, B \to C, A\}$: $\Gamma \vdash A$ (Ax), $\Gamma \vdash A \to B$ (Ax) $\Rightarrow \Gamma \vdash B$ ($\to$E); with $\Gamma \vdash B \to C$: $\Gamma \vdash C$ ($\to$E); then $\to$I three times. Curry–Howard term: $\lambda f.\lambda g.\lambda x.g(fx)$.

5 Planning: STRIPS

World fully described by its **state**; **actions** move between states; planning = find a sequence of actions from initial to goal state (same vocabulary as RL). STRIPS (Stanford Research Institute Problem Solver, 1971): strategy + formal representation. Predicates and variables Capitalized, constants lowercase.

`Move(X,Y)` Pre: `At(X), Level(low)` Post: `!At(X), At(Y)`. Other monkey actions: `ClimbUp/ClimbDown(Location)`, `MoveBox(X,Y)`, `TakeBanana(Location)`. Blocks world: special block `t` (table); `MoveBlocks(BlockMoved,BlockFrom,BlockTo)` Pre: `!Table(BM), Clear(BM), Clear(BT), On(BM,BF)` Post: `On(BM,BT), !On(BM,BF), !Clear(BT), Clear(BF)`; also `MoveFromTable`, `MoveToTable`.

STRIPS solver = **backtracking**: solves any problem but may explore pointless strategies; semi-decidable. **Sussman anomaly**: goals `On(a,b), On(b,c)` interfere (achieving one undoes the other, in either order).

6 Planning as satisfiability (Kautz & Selman 1992)

Plan $\leftrightarrow$ model of a SAT problem (bigger, but SAT solvers beat theorem provers).

1. Add a **time** variable (last): `Move(a,b)`→`Move(a,b,1)`.
2. **Ground**: enumerate all argument values (monkey: 9 `Move` at each t).
3. Grounded clause → boolean: `Move(a,b,12)`→`move_a_b_12, U-1-2`.
4. Constraints: **initial** state (*every* grounded fluent at $t = 0$, positive or negated $\Rightarrow$ one model), **goal** at $t = N$ (only goal facts $\Rightarrow$ many models), **problem constraints** (e.g. table always clear, no block on itself, exactly one action per step), **explanatory frame axioms**, **actions**.

Action: $a_t \Rightarrow (p_t \wedge c_{t+1})$ (if not taken, p_t, c_{t+1} free).

Frame axiom: a change of a state variable between t and $t+1$ implies the disjunction of the actions that cause it at t, *both directions*: $(\neg f_t \wedge f_{t+1}) \Rightarrow \bigvee\{a : a \text{ adds } f\}$, $(f_t \wedge \neg f_{t+1}) \Rightarrow \bigvee\{a : a \text{ deletes } f\}$.

e.g. `(on-A-B-5 ∧ ¬on-A-B-4)` $\Rightarrow$ `(moveFromTable-A-T-B-4 ∨ moveBlocks-A-C-B-4)`.

SAT always decidable but the planner fixes N: **increase** N until satisfiable. Lab: `Move(X,Y,Z,T)`→`Object(X,T), Source(Y,T), Destination(Z,T)` (less blow-up). Bugs usually come from *too few* constraints.

(PO)MDP: stochastic transitions $p(s'|s, a)$; reward $s \xrightarrow{a,r} s'$; compute a **policy** $\pi : s \mapsto a$ maximizing $\sum_{t=0}^{\infty} \gamma^t E[r_t]$ (expectation over $p(s_{t+1}|s_t, a_t)$ following π).

7 Formal Concept Analysis (FCA)

Formal context $K = (X, Y, I)$: objects X (rows), attributes Y (columns), $I \subseteq X \times Y$, $(x, y) \in I$: "x has y".

Derivation operators: $A^\uparrow = \{y \in Y \mid \forall x \in A : (x, y) \in I\}$ (common attributes); $B^\downarrow = \{x \in X \mid \forall y \in B : (x, y) \in I\}$ (objects having all of B). $\emptyset^\uparrow = Y$, $\emptyset^\downarrow = X$.

Formal concept (A, B): $A^\uparrow = B$ **and** $B^\downarrow = A$; A **extent**, B **intent** (perfect, maximal pairing).

Galois connection: antitone $A_1 \subseteq A_2 \Rightarrow A_2^\uparrow \subseteq A_1^\uparrow$, $B_1 \subseteq B_2 \Rightarrow B_2^\downarrow \subseteq B_1^\downarrow$; $A \subseteq A^{\uparrow\downarrow}$, $B \subseteq B^{\downarrow\uparrow}$.

Closure operator T: extensive $A \subseteq T(A)$, monotone, idempotent $T(T(A)) = T(A)$. $(\cdot)^{\uparrow\downarrow}$ closes extents, $(\cdot)^{\downarrow\uparrow}$ closes intents. Concepts = closed sets. **Every** $(B^\downarrow, B^{\downarrow\uparrow})$ **and** $(A^{\uparrow\downarrow}, A^\uparrow)$ **is a concept.**

Order: $(A_1, B_1) \preceq (A_2, B_2) \iff A_1 \subseteq A_2 \iff B_2 \subseteq B_1$ (subconcept: fewer objects, more attributes). $(\mathcal{C}(K), \preceq)$ = complete lattice = **concept lattice** (Hasse diagram).

Meet $\bigwedge_t (X_t, Y_t) = \left(\bigcap_t X_t, (\bigcup_t Y_t)^{\downarrow\uparrow}\right)$; **Join** $\bigvee_t (X_t, Y_t) = \left((\bigcup_t X_t)^{\uparrow\downarrow}, \bigcap_t Y_t\right)$.

Intersection method (all concepts by hand). Intersection of intents is an intent (join formula). So: **1** write each object intent $\{x\}^\uparrow$; **2** add Y (intent of the bottom, extent $Y^\downarrow$, often $\emptyset$); **3** repeatedly add pairwise intersections of the intents in the list until nothing new (include $X^\uparrow$ = top intent); **4** for each intent B compute extent $B^\downarrow$; **5** order by inclusion of extents, draw edges only to *immediate* neighbours. Dually: intersect the attribute extents $\{y\}^\downarrow$ (meet formula) and close with $^\uparrow$. Check: #concepts same both ways.

Reduced labelling: attribute y written at $(\{y\}^\downarrow, \{y\}^{\downarrow\uparrow})$ (highest concept containing it); object x at $(\{x\}^{\uparrow\downarrow}, \{x\}^\uparrow)$ (lowest). Intent of a node = all attributes on/above it; extent = all objects on/below it.

NextClosure (Ganter): generate closed intents one by one in **lectic order** ($y_1 < y_2 < \ldots$); from current intent B, try attributes from the last (y_n) downwards: for $y_i \notin B$, $C = ((B \cap \{y_1..y_{i-1}\}) \cup \{y_i\})^{\downarrow\uparrow}$; accept iff C adds no attribute $< y_i$ not already in B. Start $\emptyset^{\downarrow\uparrow}$, stop at Y. $O(|X||Y|^2|C|)$, small memory. Others: Close-by-One (canonicity test, $O(|X|^2|Y||C|)$), Titanic (iceberg, support), Bordat ($O(|X||Y|^2|C|)$), Chein (intersect intents in layers, $O(|X|^3|Y||C|)$), Lindig (upper neighbours, bottom-up).

Slide ex. $x_1 : y_1y_2y_3y_4$, $x_2 : y_1y_3y_4$, $x_3, x_4 : y_2y_3y_4$, $x_5 : y_1$. NextClosure: $\emptyset \to \{y_3, y_4\}$ $(\{y_4\}^{\downarrow\uparrow} = \{y_3, y_4\}$ adds earlier y_3: reject; try y_3 ok) $\to \{y_2, y_3, y_4\} \to \{y_1\} \to \{y_1, y_3, y_4\} \to Y$. 6 concepts: $C_1(X, \emptyset)$, $C_2(x_1x_2x_3x_4, y_3y_4)$, $C_3(x_1x_3x_4, y_2y_3y_4)$, $C_4(x_1x_2x_5, y_1)$, $C_5(x_1x_2, y_1y_3y_4)$, $C_6(x_1, Y)$.

Clarification / reduction: **clarified** = no identical rows (objects) and no identical columns (attributes); remove duplicates $\Rightarrow$ *isomorphic* lattice. **Reduced** = no reducible object/attribute (derivable from others, e.g. a column = intersection of other columns, or a full column); reduced $\Rightarrow$ clarified, not conversely.

Attribute implications $Z_1 \Rightarrow Z_2$ ($Z_1, Z_2 \subseteq Y$): every object having all of Z_1 has all of Z_2. **Valid** iff $Z_1^\downarrow \subseteq Z_2^\downarrow \iff Z_2 \subseteq Z_1^{\downarrow\uparrow}$.

Counterexample = object in $Z_1^\downarrow \setminus Z_2^\downarrow$. **Armstrong** (sound & complete): reflexivity ($Z_2 \subseteq Z_1 \Rightarrow Z_1 \Rightarrow Z_2$), augmentation ($Z_1 \Rightarrow Z_2 \vdash Z_1 \cup Z_3 \Rightarrow Z_2 \cup Z_3$), transitivity. **Non-redundant basis**: complete + no member derivable from others. **Duquenne–Guigues basis**: premises = **pseudo-intents** (non-closed sets), each $P \Rightarrow P^{\downarrow\uparrow}$; minimum number of implications.

Extensions: many-valued context (X, Y, W, I), scaling, pattern structures, icebergs, stability, triadic/polyadic, relational CA.

8 Association rules

Itemset; **support count** $\sigma(X) = \#$transactions containing X; **support** $s(X) = \sigma(X)/|T|$; frequent: $s(X) \geq minSup$. Rule $X \Rightarrow Y$ (non-empty): $s(X \Rightarrow Y) = \frac{\sigma(X \cup Y)}{|T|}$; **confidence** $c(X \Rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$ $(= P(Y|X))$. Co-occurrence, NOT causality.

Brute force: 2^d itemsets, $R = \sum_{i=1}^{d-1} C_d^i \sum_{j=1}^{d-i} C_{d-i}^j$ rules. Cost = #**passes** over the DB.

A priori principle: frequent $\Rightarrow$ all subsets frequent; infrequent $\Rightarrow$ all supersets infrequent (support anti-monotone: $X \subseteq Y \Rightarrow s(X) \geq s(Y)$). **Apriori** (Agrawal & Srikant 1994), level-wise: L_1 = frequent items; pass k: *join* L_{k-1} to candidates C_k, *prune* candidates with an infrequent $(k-1)$-subset, scan, keep $\geq minSup$; stop when none.

Rules from frequent L: for each non-empty $A \subset L$: $A \Rightarrow L \setminus A$, $c = s(L)/s(A)$, keep $\geq minConf$.

Lift $= \frac{c(A \Rightarrow Y)}{s(Y)} = \frac{s(X \cup Y)}{s(X)s(Y)}$: $= 1$ independent, > 1 positive correlation (interesting), < 1 negative. Tea⇒Coffee $c = 75\%$ but s(coffee) = 80%: lift < 1. Others: leverage, conviction, Zhang, Jaccard, certainty, Kulczynski.

Algo	passes	idea
Apriori	k	level-wise prune (anti-monotone)
PCY	2	hash pairs in pass 1 (memory)
SON	2	chunks in parallel, verify globally
Toivonen	1 (or 2)	sample + negative border
FP-Growth	2	FP-tree (prefix merge), no candidates

FP-max: maximal frequent itemsets only.

FCA vs AR: FCA implication $Z_1 \Rightarrow Z_2$ = deterministic, holds for *all* objects; AR $X \Rightarrow Y$ = probabilistic, measured by support/confidence/lift. **Implication = AR with confidence 100%**: valid $\iff Z_1^\downarrow \subseteq Z_2^\downarrow \iff \sigma(Z_1 \cup Z_2) = \sigma(Z_1) \iff c = 1$. Closed itemsets = concept intents ($s(B) = |B^\downarrow|/|X|$). FCA: small, clean, exact knowledge (lattice, DG basis). AR: large noisy data, thresholds, scalable algorithms.

9 Fuzzy logic

Probability vs degree of truth: differ in (1) semantics, (2) aggregation. Cooks 60/60/50/50%: P(some cooks)$= 1-.4\cdot.4\cdot.5\cdot.5 = 96\%$; degree of "great food" = 60% (max).

Negation $\neg x = 1 - x$; **Yager** $\neg_w(x) = (1 - x^w)^{1/w}$ ($w = 1$ standard).

Average $\sum_i a_i/n$ (not sensitive to small values); harmonic mean $n/\sum_i 1/a_i$ (sensitive, one small value cannot annihilate). Not associative $\Rightarrow$ apply to all conjuncts at once.

T-norm $T : [0,1]^2 \to [0,1]$: commutative, monotone ($T(a, c) \leq T(b, c)$ if $a \leq b$), associative, neutral $T(a, 1) = a \Rightarrow T(a, 0) = 0$. Used for $\wedge$.

$\min$ $T = \min(a, b)$ (largest); product $a \times b$; Łukasiewicz $\max(0, a + b - 1)$; drastic (a=1)?b : (b=1)?a : 0 (smallest). Weak law of contradiction $T(a, 1 - a) \leq 0.5$.

T-conorm $\perp(a, b) = 1 - T(1 - a, 1 - b)$: comm., monotone, assoc., identity $\perp(a, 0) = a$; for $\vee$. max (smallest), probabilistic sum $1-(1-a)(1-b)$, bounded sum $\min(a+b, 1)$, drastic a=0?b : b=0?a : 1 (largest).

S-implication = value of $\neg x \vee y$: $\max(1 - x, y)$; $1 - x(1 - y)$; $\min(1 - x + y, 1)$. Can be < 1 even if $x < y$!

R-implication / residuum $R_T(x, y) = \sup\{z : T(z, x) \leq y\}$: $= 1$ if $x \leq y$, else min: y; product: y/x; Łuk: $1 - x + y$. Implication **satisfied** if $x \leq y$. Equals crisp $\supset$ on {0, 1}, no modus tollens.

Interpretation I: $I(\neg a) = 1 - a$, $I(a \wedge b) = T(a, b)$, $I(a \vee b) = 1 - T(1 - a, 1 - b)$, $I(a \Rightarrow b) = R_T$. Slide ex. (product): $I((A \Rightarrow \neg G) \wedge A) = I(.22 \wedge .9) = 0.198$.

Tasks: (1) **interpretation** (formula value from statements); (2) **deduction/inference**.

Simple positive fuzzy inference problem : rules $P_1 \wedge .. \wedge P_n \Rightarrow_\phi C$, ϕ conjunctive aggregation; satisfied if $C \geq \phi(P_1..P_n)$; **solution**: (1) values $\geq$ initial, (2) all rules satisfied, (3) no output can be smaller. **Theorem**: iterate satisfying rules in any order until all satisfied (if ϕ monotonic, continuous). Negations $\Rightarrow$ may have no solution / oscillate.

Fuzzy set $\mu : U \to [0, 1]$; support $\{x : \mu(x) > 0\}$; core $\{x : \mu(x) = 1\}$; α-cut $\{x : \mu(x) \geq \alpha\}$; $\cap/\cup$ = pointwise T-norm/conorm; α-capping $\mu_{\leq\alpha}(x) = \min(\alpha, \mu(x))$. **Equality relation** $E : X^2 \to [0, 1]$: commutative, reflexive $E(x, x) = 1$, transitive $T(E(x, y), E(y, z)) \leq E(x, z)$. **Fuzzification** of crisp A: $\mu(x) = \sup\{E(x, y) : y \in A\}$.

Defuzzification (fuzzy set → element): **First of Maxima** $\min\{c \mid \forall x : \mu(c) \geq \mu(x)\}$ – *safe choice* (is a max) but ignores other maxima; **Mean of Maxima** $avg\{\ldots\}$ – may land on a non-maximum; **Median** $\sum_{x' < x} \mu(x') = \sum_{x' > x} \mu(x')$ – may also land on a non-max.

Mamdani p_1 is $P_1 \wedge .. \Rightarrow c$ is C: 1 fuzzify inputs, 2 firing strength $\alpha = T(\ldots)$, 3 cap C at α, 4 defuzzify: $c = defuzz(C_{\leq T(P_1(p_1),\ldots)})$. Ex: HOT .9, CROWDED .6 $\Rightarrow \alpha = \min = .6$.

10 MAX SAT & Probabilistic Soft Logic

Contradictory formulas: MAX SAT (maximize #satisfied). **Weighted MAX SAT**: maximize sum of weights of satisfied formulas. Generalizes MAX SAT $\supseteq$ SAT (NP-complete) $\Rightarrow$ in general no faster than trying all assignments (not poly unless P=NP).
Ex: $\{s\}[7]$, $\{\neg s, \neg b\}[8]$, $\{b\}[10]$: worlds give 18 (s=0, b=1), 15, 17, 10 $\rightarrow$ best 18.
Relaxation (Łukasiewicz, S-implication): clause $\{x_1..x_n\}$ has degree $\min(\sum_i x_i, 1)$; with $y : S \rightarrow [0,1]$: $\min(\sum_{s+} y(s) + \sum_{s-} (1 - y(s)), 1)$. Ex $s = .3, b = .7$: $.3 \cdot 7 + 1 \cdot 8 + .7 \cdot 10 = 17.1$.
PSL inference: find $y : S \rightarrow [0,1]$ maximizing $\sum_{c \in C} w(c) \min(\sum_{s+ \in c} y(s) + \sum_{s- \in c} (1 - y(s)), 1)$ s.t. $y(s) \leq v$.
Convex $\Rightarrow$ polynomial ("hit and run"): start feasible, pick a variable, optimize it, repeat until convergence). Statements fuzzy, formulas weighted.

11 Theorem proving & λ-calculus

Tools: LEAN (Microsoft), Isabelle, Coq/Rocq. **Hilbert's program** (1920s): finite axioms, **consistent** (no contradiction), **complete** (P or $\neg P$ provable), **decidable** (program decides). **Gödel 1**: any consistent formal system containing arithmetic is incomplete $\Rightarrow$ infeasible. **Gödel 2**: it cannot prove its own consistency. Mathlib $\sim$275k theorems.
λ-terms : variables; abstraction $\lambda x.y$ ($x \mapsto y$); application fx (left assoc.: $fxy = (fx)y$). Untyped. **β-reduction** $(\lambda x.M)N \rightarrow M[N/x]$: $(\lambda x.xx)y \rightarrow yy$, $(\lambda x.\lambda y.x)z \rightarrow \lambda y.z$ (rename bound vars to avoid capture). **Normal form**: cannot reduce. **Strongly normalizing**: every reduction sequence terminates. Counter-ex $(\lambda x.xx)(\lambda x.xx)$ (cannot be typed).

0	$\lambda f.\lambda x.x$	$n+1$	$\lambda n.\lambda f.\lambda x.f(nfx)$
1	$\lambda f.\lambda x.fx$	$m+n$	$\lambda m.\lambda n.\lambda f.\lambda x.mf(nfx)$
n	$\lambda f.\lambda x.f^n x$	mn	$\lambda m.\lambda n.\lambda f.\lambda x.m(nf)x$
True	$\lambda a.\lambda b.a$	$\neg$	$\lambda p.p$ False True
False	$\lambda a.\lambda b.b$	$\wedge$ / $\vee$	$\lambda p.\lambda q.pqp$ / $\lambda p.\lambda q.ppq$

$1 + 2$: $(\lambda n.\lambda m.\lambda f.\lambda x.nf(mfx))(1)(2) \rightarrow \lambda m.\lambda f.\lambda x.f(mfx)\,(2) \rightarrow \lambda f.\lambda x.f(f(fx)) = 3$.
Simply typed : $x : \sigma$; $\lambda x.y : \sigma \rightarrow \tau$ if $x : \sigma, y : \tau$; $fx : \tau$ only if $f : \sigma \rightarrow \tau$, $x : \sigma$. **Strong normalization thm** (Curry): simply typed λ-calculus is strongly normalizing.
Curry–Howard : proofs $\leftrightarrow$ programs; formula = type, proof = term. $\frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightarrow B}$ ($\rightarrow$I) $\leftrightarrow$ $\frac{\Gamma, x:\sigma \vdash y:\tau}{\Gamma \vdash \lambda(x:\sigma).y:\sigma \rightarrow \tau}$ (Abs); $\rightarrow$E (modus ponens) $\leftrightarrow$ App $\frac{\Gamma \vdash f:\sigma \rightarrow \tau \quad \Gamma \vdash x:\sigma}{\Gamma \vdash fx:\tau}$.
$\rightarrow$ function, $\wedge$ product $\sigma \times \tau$, $\vee$ sum $\sigma + \tau$, $\top$ unit 1, $\bot$ empty 0, $\forall$ dependent product, $\exists$ dependent sum. Still strongly normalizing.
LEAN: dependent type theory; types Bool, Type$\rightarrow$Type, $\times$, $\oplus$, inductive Nat. Tactics: `rfl, rw, intro, exact, apply, induction, by_contra, linarith, ring, simp`.
Non-constructive proof: a^b rational with a, b irrational via $x = \sqrt{2}^{\sqrt{2}}$ (case split) uses **excluded middle** $P \vee \neg P$ (classical axiom). $\sqrt{2}$ irrational: $a^2 = 2b^2 \Rightarrow a$ even $\Rightarrow b$ even, contradicts coprime.

λ exercise sheet: (1) $(\lambda x.x)y \rightarrow y$; (2) $(\lambda x.yx)z \rightarrow yz$; (3) $(\lambda x.\lambda y.x)(\lambda x.y) \rightarrow \lambda w.\lambda x.y$ (rename bound y to w: else capture); (4) $(\lambda x.xy)(\lambda z.z) \rightarrow (\lambda z.z)y \rightarrow y$; (5) $(\lambda x.\lambda y.xy(xy))(\lambda z.zz) \rightarrow \lambda y.(\lambda z.zz)y(\lambda z.zz)y) \rightarrow \lambda y.yy(yy)$. Ex2 $\lambda n.n(\lambda x.\text{false})$true: **is zero?** (0 returns true; $n \geq 1$ applies the constant false). Ex3 $\lambda m.\lambda n.nm$ = **exponentiation** m^n (n applies "$\times m$" n times). Ex4 predecessor: n iterates $\lambda g.\lambda h.h(gf)$ starting from $\lambda u.x$ (which swallows the first f), $\lambda u.u$ extracts $\Rightarrow f^{n-1}x$.

12 Description logics (DL)

Family of logic-based KR languages; **variable-free** syntax, readable; most DLs = **decidable fragments of FOL** (FOL expressive but undecidable). Atomic concepts (classes) A, B, atomic roles R, individuals.
KB = TBox (terminology: concepts + roles) **+ ABox** (assertions on named individuals). e.g. TBox Mother $\equiv$ Woman $\sqcap \exists$hasChild.$\top$; ABox Woman(Alice), hasChild(Alice,Bob) $\Rightarrow$ Mother(Alice).
$\mathcal{AL}$: $\top$, $\bot$, $\neg A$ (atomic only), $C \sqcap D$, $\forall R.C$ (value restriction), $\exists R.\top$ (limited $\exists$). $\mathcal{U}$: $C \sqcup D$; $\mathcal{E}$: $\exists R.C$; $\mathcal{C}$: $\neg C$ (full); $\mathcal{N}$: $\leq nR$, $\geq nR$. $\mathcal{AL} \subseteq \mathcal{ALE} \subseteq \mathcal{ALC} \subseteq \mathcal{SHOIN} \subseteq \mathcal{SROIQ}$ (OWL 2 DL $= \mathcal{SROIQ}(D)$). Qualified: $\geq$ 2hasChild.Professor. Inverse, transitive roles, subroles, concrete domains.
Person $\sqcap \forall$hasChild.$\bot$ = no children; $\exists$hasChild.$\top$ = at least one child; Person $\sqcap$ Female $\sqcap \geq$ 3hasChild.
Semantics $\mathcal{I} = (\Delta^I, \cdot^I)$: $A^I \subseteq \Delta^I$, $R^I \subseteq \Delta^I \times \Delta^I$; $(C \sqcap D)^I = C^I \cap D^I$, $(C \sqcup D)^I = C^I \cup D^I$, $(\neg C)^I = \Delta^I \setminus C^I$, $(\forall R.C)^I = \{x \mid \forall y, (x,y) \in R^I \rightarrow y \in C^I\}$, $(\exists R.C)^I = \{x \mid \exists y, (x,y) \in R^I \wedge y \in C^I\}$, $\top^I = \Delta^I$, $\bot^I = \emptyset$.

DL	FOL (free var x)
$C \sqcap D$ / $C \sqcup D$ / $\neg C$	$C(x) \wedge D(x)$ / $C(x) \vee D(x)$ / $\neg C(x)$
$\exists R.C$	$\exists y.R(x,y) \wedge C(y)$
$\forall R.C$	$\forall y.R(x,y) \rightarrow C(y)$
$C \sqsubseteq D$	$\forall x.C(x) \rightarrow D(x)$
$R_1 \circ R_2 \sqsubseteq R_3$	$\forall x,y,z.R_1(x,y) \wedge R_2(y,z) \rightarrow R_3(x,z)$

e.g. Parent $\equiv$ Person $\sqcap \exists$hasChild.Person: $\forall x.Parent(x) \leftrightarrow (Person(x) \wedge \exists y.(hasChild(x,y) \wedge Person(y)))$.
TBox axioms: inclusion $C \sqsubseteq D$, equivalence $C \equiv D$; $\mathcal{I} \models \mathcal{T}$ (**model**) iff it satisfies every axiom. **ABox**: $C(a)$ ($a^I \in C^I$), $R(b,c)$ ($(b^I, c^I) \in R^I$, c is a filler of R for b). Model of $\mathcal{A}$ w.r.t. $\mathcal{T}$: model of both.
TBox reasoning : **satisfiability** ($C^I \neq \emptyset$ in some model of $\mathcal{T}$); **subsumption** $\mathcal{T} \models C \sqsubseteq D$ ($C^I \subseteq D^I$ all models); **equivalence** $C \equiv_{\mathcal{T}} D$; **disjointness** $C^I \cap D^I = \emptyset$.
Reductions: C unsat $\iff$ $C \sqsubseteq \bot$; equiv $\iff$ both $\sqsubseteq$; disjoint $\iff$ $C \sqcap D \sqsubseteq \bot$. $C \sqsubseteq D$ $\iff$ $C \sqcap \neg D$ unsat; equiv $\iff$ $C \sqcap \neg D$ and $\neg C \sqcap D$ unsat; disjoint $\iff$ $C \sqcap D$ unsat.
ABox reasoning : **consistency** (exists model of $\mathcal{A}$ and $\mathcal{T}$); **instance check** $\mathcal{A} \models C(a)$ iff $\mathcal{A} \cup \{\neg C(a)\}$ inconsistent; **retrieval** (all a with $\mathcal{A} \models C(a)$); **realization** (most specific concepts of a, minimal w.r.t. $\sqsubseteq$).

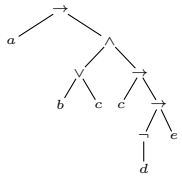
Ex. $\mathcal{T}$ = {Father$\sqsubseteq$Man, Mother$\sqsubseteq$Woman, Man$\sqcap$Woman$\sqsubseteq \bot$}, $\mathcal{A}$ = {Mother(mary), Father(mary)}: $mary^I \in Mother^I \subseteq Woman^I$ and $\in Father^I \subseteq Man^I$, but $Man^I \cap Woman^I = \emptyset \Rightarrow$ **inconsistent** w.r.t. $\mathcal{T}$ (consistent w.r.t. empty TBox).

Exam-style DL (old course slides, Bloch). *Translation* τ_x: $\tau_x(C) = C(x)$, $\tau_x(C \sqcap D) = \tau_x(C) \wedge \tau_x(D)$, $\sqcup \rightarrow \vee$, $\tau_x(\exists r.C) = \exists y, r(x,y) \wedge \tau_y(C)$, $\tau_x(\forall r.C) = \forall y, r(x,y) \rightarrow \tau_y(C)$; TBox $\bigwedge_{C \sqsubseteq D} \forall x(\tau_x(C) \rightarrow \tau_x(D))$; ABox $a : C \rightarrow C(a)$, $(a,b) : r \rightarrow r(a,b)$. Also: value restr. $\ni r.\{a\}$, $(\geq nR)^I = \{x \mid |\{y \mid (x,y) \in R^I\}| \geq n\}$; "only humans have human children" $\exists$hasChild.Human$\sqsubseteq$Human.
(1) Prove $\forall r.(A \sqcap B) \sqsubseteq \forall r.A \sqcap \forall r.B$. *Interpretations*: take any $\mathcal{I}$, $x \in (\forall r.(A \sqcap B))^I$; for every y with $(x,y) \in r^I$: $y \in A^I \cap B^I \subseteq A^I$, so $x \in (\forall r.A)^I$; same for B; hence $x \in (\forall r.A)^I \cap (\forall r.B)^I$. *FOL:* $\forall x[\forall y(r(x,y) \rightarrow A(y) \wedge B(y)) \rightarrow (\forall y(r(x,y) \rightarrow A(y)) \wedge \forall y(r(x,y) \rightarrow B(y)))]$ is valid (or: negate, skolemize, resolution $\rightarrow$ []). Converse also holds ($\equiv$). Careful: $\exists r.(A \sqcap B) \sqsubseteq \exists r.A \sqcap \exists r.B$ but *not* conversely (two different fillers).
(2) Interpretation: Father$\equiv$ $\neg$Female$\sqcap\exists$hasChild.Human, Δ^I = $\{John, Mary\}$, Human^I = $\{John, Mary\}$, hasChild^I = $\{(John, Mary)\} \Rightarrow (\exists$hasChild.Human$)^I = \{John\}$, Father$^I = \{John\}$.
(3) Instance check TBox: Female, Child, Works, StudiesAtUni $\sqsubseteq$ Human; SuccessfulMan $\equiv \neg$Female$\sqcap$InBusiness$\sqcap\exists$married.Lawyer$\sqcap\exists$child.(StudiesAtUni$\sqcup$Works). ABox: Pedro:$\neg$Female, Pedro:InBusiness, Mary:Lawyer, John:Works, (Pedro,Mary):married, (Pedro,John):child. *Is Pedro a SuccessfulMan?* **Yes**: Mary witnesses $\exists$married.Lawyer; John$\in$Works$\subseteq$StudiesAtUni$\sqcup$Works witnesses $\exists$child.(...); with $\neg$Female, InBusiness all conjuncts hold in every model, and $\equiv$ (sufficient condition) gives $\mathcal{K} \models$ Pedro:SuccessfulMan. (With only $\sqsubseteq$ it would NOT follow.) Method: list each conjunct, give a witness for each $\exists$, check all fillers for each $\forall$.
(4) Open world: from MotherWithoutDaughter(Mary) $\equiv$ Mother$\sqcap\forall$hasChild.$\neg$Woman and hasChild(Mary,Peter): Peter: $\neg$Woman. A $\forall$ only constrains *existing* fillers; absence of an assertion $\neq$ its negation.

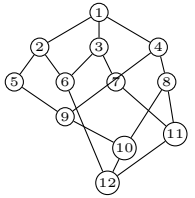
Ontology : "explicit formal specification of the terms in the domain and relations among them" (Gruber 93); "formal, explicit specification of a shared conceptualization" (Hudelot 06). Structure $O := \{C, R, H^C, rel, A^O\}$ ($H^C \subseteq C \times C$ taxonomy, $rel : R \rightarrow C \times C$, dom, range, axioms). KB := $\{O, I, inst, instr\}$: Types (Guarino 98): top-level, domain, task, application. Lightweight (E-R, UML) vs heavyweight (DL, FOL). KG = instance level (ABox, triples s–p–o); ontology = schema (TBox).
OWL (on RDF/RDFS; Lite $\subset$ DL $\subset$ Full): restrictions $\exists$ someValuesFrom, $\forall$ allValuesFrom, $\ni$ hasValue, $=/\leq/\geq$ cardinality. owl:Thing top, owl:Nothing bottom. Property characteristics: inverse $P_1(X,Y) \iff P_2(Y,X)$; transitive; symmetric; **functional** $P(X,Y) \wedge P(X,Z) \Rightarrow Y = Z$; **inverse functional** $P(Y,X) \wedge P(Z,X) \Rightarrow Y = Z$. Reasoners HermiT, Pellet, FaCT++; Protégé: EquivalentTo = necessary & sufficient; SubClassOf = necessary.
Rule reasoning : modus ponens $\frac{A,\ A \rightarrow B}{B}$, modus tollens $\frac{\neg B,\ A \rightarrow B}{\neg A}$. Cycle: detection (pattern matching/unification) $\rightarrow$ choice (first rule as Prolog, most specific, most useful; metarules) $\rightarrow$ execution. **Forward** (data driven) vs **backward** (goal oriented, recursive on premises) vs mixed. West is criminal: American(x)$\wedge$Arm(y)$\wedge$Sells(x,y,z)$\wedge$Hostile(z)$\Rightarrow$Criminal(x), $\{x/West, y/M1, z/Nono\}$.

13 Example exam – brief solutions

PL $F = a \rightarrow ((b \vee c) \wedge (c \rightarrow (\neg d \rightarrow e)))$. False iff $a = 1$ and $[b = c = 0$ (d, e any) or $c = 1, d = 0, e = 0$ (b any)]: $(a,b,c,d,e) \in \{10000, 10001, 10010, 10011, 10100, 11100\}$ (6). CNF: $\langle[\neg a, b, c], [\neg a, \neg c, d, e]\rangle$ (via $\neg a \vee (\ldots)$ distributed; $c \rightarrow (\neg d \rightarrow e) \equiv \neg c \vee d \vee e$).
Decomposition tree (root = main connective):



FOL $G = (a(x,y) \rightarrow \forall x.\exists y.(\neg b(x,y,c(z,y)) \vee d(x))) \wedge e(f(x,g(y)))$. Predicates: $a/2$, $b/3$, $d/1$, $e/1$ (formula positions). Functions: $c/2$, $f/2$, $g/1$ (inside terms). Free: x, y (in $a(x,y)$, $e(f(x,g(y)))$) and z (never quantified) $\Rightarrow \{x,y,z\}$.
Planning (vars `posX-i-t`). Init: `posX-1-0` $\wedge \neg$`posX-2-0` $\wedge \neg$`posX-3-0` $\wedge$ `posY-1-0` $\wedge \neg$`posY-2-0` $\wedge \neg$`posY-3-0`. Goal: `posX-2-5` $\wedge$ `posY-3-5`. Frame, posX at $t = 3$:
$(\neg pX1_3 \wedge pX1_4) \rightarrow L2_3$; $(pX1_3 \wedge \neg pX1_4) \rightarrow R1_3$;
$(\neg pX2_3 \wedge pX2_4) \rightarrow R1_3 \vee L3_3$; $(pX2_3 \wedge \neg pX2_4) \rightarrow R2_3 \vee L2_3$;
$(\neg pX3_3 \wedge pX3_4) \rightarrow R2_3$; $(pX3_3 \wedge \neg pX3_4) \rightarrow L3_3$ ($L2_3$=`L-2-3`).
Up at $t = 2$: `U-1-2` $\rightarrow$ `posY-1-2` $\wedge \neg$`posY-1-3` $\wedge$ `posY-2-3`; `U-2-2` $\rightarrow$ `posY-2-2` $\wedge \neg$`posY-2-3` $\wedge$ `posY-3-3`; $\neg$`U-3-2` (pre $Y < 3$ false).
Fuzzy Average $T(a,b) = \frac{a+b}{2}$: **not** a T-norm: $T(0,1) = 0.5 \neq 0$ (neutral element fails); also not assoc.: $T(T(0,0),1) = .5 \neq T(0,T(0,1)) = .25$. $(a \rightarrow b) \vee c$, R-impl. of min, $a = .3 \leq b = .7 \Rightarrow 1$; $\vee$ = max $\Rightarrow \max(1,.1) = 1$. Inference $a \wedge b \rightarrow c$, $c \rightarrow a$, $b = .3, c = .1$ ($a = 0$): rule 2 $\Rightarrow a = .1$; rule 1: $c \geq \min(.1,.3) = .1$ ok $\Rightarrow (a,b,c) = (.1,.3,.1)$. Defuzz: first of maxima (safe, a true max; ignores other maxima). W.MAX SAT: NP-hard (generalizes SAT) $\Rightarrow$ not poly in general. With $[0,1]$ degrees (Łuk.): PSL, convex $\Rightarrow$ poly (hit and run). PSL $\{a\}[1], \{\neg a, \neg b\}[2], \{b\}[3]$: maximize $a + 2\min(2 - a - b, 1) + 3b$; for $a + b \leq 1$: $a + 2 + 3b$, else $4 - a + b$ $\Rightarrow a = 0, b = 1$, value 5.
FCA (ME Middle-earth, W, M Magic, I Immortal, R Ring, Fe, E). Rows: Fr{ME,R,Fe}, Ar=Bo{ME,W,Fe}, Ga{ME,W,M,I,Fe}, Le{ME,W,I,Fe}, Sa{ME,M,I,R,E}, Go{ME,R,E}. **Clarification needed**: Aragorn = Boromir (identical rows); ME is a full column (in every intent, reducible). Intersections give 12 intents:



1 ME	all 7
2 ME,Fe	Fr,Ar,Ga,Le,Bo
3 ME,R	Fr,Sa,Go
4 ME,I	Ga,Le,Sa
5 ME,W,Fe	Ar,Ga,Le,Bo
6 ME,R,Fe	Fr
7 ME,R,E	Sa,Go
8 ME,M,I	Ga,Sa
9 ME,W,I,Fe	Ga,Le
10 ME,W,M,I,Fe	Ga
11 ME,M,I,R,E	Sa
12 Y	$\emptyset$

(a) $\{Fe\}^{\downarrow} = \{Fr, Ar, Ga, Le, Bo\}$, $^{\uparrow} = \{ME, Fe\} \rightarrow$ c2. (b) $\{M\}^{\downarrow} = \{Ga, Sa\}$, $^{\uparrow} = \{ME, M, I\} \rightarrow$ c8. (c) $\{W\}^{\downarrow} = \{Ar, Ga, Le, Bo\}$, $^{\uparrow} = \{ME, W, Fe\} \rightarrow$ c5. Verify: $(\{Ga, Sa\}, \{M, I\})$ **no**: $\{Ga, Sa\}^{\uparrow} = \{ME, M, I\}$. $(\{Fr, Ar\}, \{Fe\})$ **no**: $\{Fe\}^{\downarrow}$ has 5 objects. Implications: $\{W, Fe\} \rightarrow \{I\}$ **invalid**, counterex. Aragorn, Boromir; $s = 2/7$, $c = 2/4 = 0.5$. $\{R, Fe\} \rightarrow \{\neg E\}$: only Frodo, not Evil $\Rightarrow$ **holds**; $s = 1/7$, $c = 1$. AR link: e.g. $\{W\} \Rightarrow \{Fe\}$ $c = 4/4 =$ implication; $\{Fe\} \Rightarrow \{W\}$ $c = 4/5 =$ AR only.

14 Proof templates (past-exam style)

Galois (write $\alpha = \cdot^{\uparrow}$, $\beta = \cdot^{\downarrow}$): $Y \subseteq \alpha(X) \iff X \subseteq \beta(Y)$ (both say $X \times Y \subseteq I$). *Antitone*: $X \subseteq X', m \in \alpha(X') \Rightarrow m$ shared by all of $X' \supseteq X \Rightarrow m \in \alpha(X)$. *Extensive*: $\alpha(X) \subseteq \alpha(X) \Rightarrow X \subseteq \beta\alpha(X)$. $\alpha\beta\alpha = \alpha$: $\supseteq$ by extensivity on $\alpha(X)$; $\subseteq$ by antitone applied to $X \subseteq \beta\alpha(X)$. *Meet is a concept*: $(X_1 \cap X_2, \alpha\beta(Y_1 \cup Y_2))$; $\beta(Y_1 \cup Y_2) = \beta(Y_1) \cap \beta(Y_2) = X_1 \cap X_2$, so it is $(\beta(Z), \alpha\beta(Z))$ with $Z = Y_1 \cup Y_2$, and $\beta\alpha\beta = \beta$.
Anti-monotone support: $Y_1 \subseteq Y_2$: every transaction containing Y_2 contains $Y_1 \Rightarrow \sigma(Y_1) \geq \sigma(Y_2) \Rightarrow$ prune supersets of infrequent sets (Apriori).
T-norm proofs: min largest: $T(a,b) \leq T(a,1) = a$ (monotone + neutral) and likewise $\leq b$. $T(a,0) \leq T(1,0) = 0$. Is f a T-norm? Check the 4 axioms; refute with one numeric counter-example (neutral $f(a,1) = a$ fails most often). T-conorm properties follow from the definition $1 - T(1-a, 1-b)$.
English$\rightarrow$FOL: "all friends of John know one of his children": $(\forall x)(friend(x,john) \supset (\exists y)(child(y,john) \wedge know(x,y)))$; prenex $(\forall x)(\exists y)(friend(x,john) \supset (child(y,john) \wedge know(x,y)))$; negation $(\exists x)(\forall y)(friend(x,john) \wedge (\neg child(y,john) \vee \neg know(x,y)))$. "Nobody smokes in cafés nor restaurants": $(\forall x)(\forall y)((cafe(y) \vee restaurant(y)) \supset \neg smokes(x,y))$. Pattern: "all A are B" $\forall x(A \supset B)$; "some A is B" $\exists x(A \wedge B)$.
Unsatisfiable KB, 2 methods: (i) truth table / case split; (ii) resolution: put all formulas in clauses, derive [].